

WINNER, SD, USA

WMO: 726518

Lat: 43.391N

Lon: 99.842W

Elev: 619

StdP: 94.10

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 94990

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.8	-18.0	-24.7	0.4	-20.1	-22.4	0.5	-17.8	15.9	-3.7	14.7	-4.3	4.7	310	0.681

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	37.0	21.6	34.3	21.4	32.5	20.9	24.2	31.6	23.3	30.7	22.3	29.9	6.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	18.2	28.3	21.1	17.0	27.5	19.9	15.8	26.6	76.4	31.4	72.4	30.5	68.7	30.0	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.0	12.1	11.0	-24.9	40.8	2.8	2.6	-26.9	42.6	-28.6	44.2	-30.2	45.6	-32.2	47.5		
(5)			-25.2	25.7	2.7	0.8	-27.1	26.3	-28.8	26.7	-30.3	27.2	-32.3	27.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	-3.8	-3.1	3.2	8.9	14.7	20.7	25.1	23.5	18.8	10.7	3.8	-3.0		
	DBStd	11.78	7.23	7.54	7.44	5.91	4.95	4.08	3.79	3.59	5.13	5.80	6.73	6.99		
	HDD10.0	1804	428	371	232	87	14	0	0	0	2	62	204	404		
	HDD18.3	3599	686	601	470	284	132	22	2	3	55	244	437	662		
	CDD10.0	1821	1	3	21	56	161	322	469	419	267	83	18	1		
	CDD18.3	573	0	0	1	3	21	94	213	164	70	7	0	0		
	CDH23.3	6422	0	1	19	66	275	1005	2466	1688	788	104	9	0		
	CDH26.7	2937	0	0	3	16	89	401	1272	786	337	32	1	0		
(14) Wind	WSAvg	5.3	5.4	5.3	5.7	5.9	5.5	4.9	4.7	4.8	5.3	5.4	5.3	5.3		
(15) Precipitation	PrecAvg	531	10	12	25	68	88	104	54	50	50	46	16	9		
	PrecMax	719	24	30	63	161	204	168	157	134	183	118	43	25		
	PrecMin	335	2	1	0	18	5	43	2	9	8	1	0	0		
	PrecStd	99	6	8	18	38	53	35	41	34	40	36	12	8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.0	20.1	26.3	28.5	32.6	36.7	41.1	38.1	36.3	30.2	24.5	15.9		
		MCWB	7.0	9.5	12.5	14.2	19.3	22.3	21.7	21.4	19.6	16.2	11.9	7.4		
	2%	DB	11.4	14.4	21.7	25.0	29.0	33.0	37.6	35.9	33.3	26.3	19.9	11.5		
		MCWB	5.1	7.0	11.0	13.8	18.3	21.6	21.9	21.5	19.4	14.6	10.4	5.2		
	5%	DB	8.0	10.6	17.8	22.1	26.4	31.0	35.3	33.0	30.8	22.8	16.3	8.3		
		MCWB	3.4	4.7	9.6	12.3	16.9	20.7	22.3	21.0	18.8	13.4	8.4	3.6		
	10%	DB	6.0	7.4	14.1	19.0	23.7	28.7	32.9	31.2	27.8	20.0	13.0	6.1		
		MCWB	2.2	3.2	7.3	10.8	15.7	20.0	21.6	20.7	17.6	12.0	6.7	2.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.4	10.3	15.0	16.2	21.1	24.6	25.7	24.8	22.4	17.8	13.0	7.8		
		MCDB	14.8	18.5	20.1	25.2	29.6	32.1	33.1	32.7	30.5	26.4	22.0	14.7		
	2%	WB	5.3	7.5	11.8	14.5	19.5	23.1	24.6	23.5	21.1	16.0	10.8	5.5		
		MCDB	10.7	13.8	19.9	23.7	26.8	30.3	32.0	31.0	29.6	23.4	18.8	10.8		
	5%	WB	3.8	5.1	9.9	12.9	18.0	21.9	23.7	22.5	20.1	14.3	9.0	3.8		
		MCDB	7.9	10.6	17.4	20.5	25.1	29.1	31.2	29.9	27.9	21.4	16.0	8.2		
	10%	WB	2.3	3.2	7.9	11.5	16.4	20.8	22.8	21.7	18.9	12.6	7.1	2.3		
		MCDB	5.6	7.0	14.1	17.8	22.5	27.4	30.5	28.9	26.2	19.4	12.9	5.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	11.3	12.0	12.4	12.1	12.5	13.4	13.0	13.4	12.4	11.8	9.8		
		MCDBR	14.0	15.5	17.6	17.7	15.7	15.2	15.8	15.7	16.2	17.0	16.1	13.1		
	5% WB	MCWBR	9.2	9.3	9.4	9.0	7.9	6.7	5.5	5.9	6.3	8.5	8.9	8.6		
		MCWBR	13.4	14.8	16.0	15.9	13.9	13.2	13.6	13.6	13.8	15.4	15.7	12.6		
(40) Clear-Sky Solar Irradiance	taub		0.261	0.273	0.296	0.341	0.357	0.372	0.391	0.393	0.349	0.314	0.276	0.263		
	taud		2.523	2.505	2.496	2.363	2.406	2.395	2.371	2.346	2.440	2.508	2.534	2.507		
	Ebn at Noon		896	939	952	924	915	898	876	862	880	875	867	856		
	Edh at Noon		67	81	94	116	115	117	118	117	99	81	66	62		
(44) All-Sky Solar Radiation	RadAvg		1.95	2.89	3.94	4.96	5.89	6.67	6.88	5.80	4.66	3.15	2.14	1.61		
	RadStd		0.15	0.21	0.33	0.34	0.54	0.47	0.26	0.41	0.31	0.36	0.19	0.12		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page